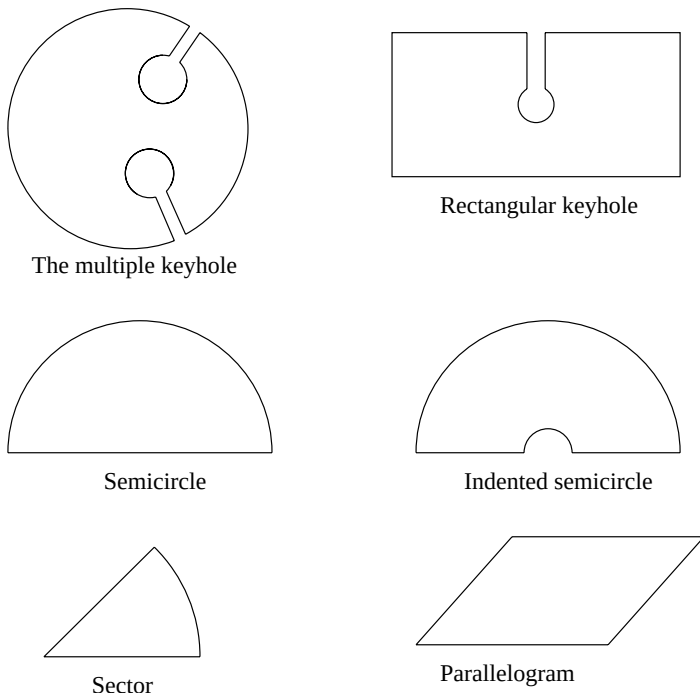


**Figure 7.** Examples of toy contours

EXAMPLE 1. We show that if $\xi \in \mathbb{R}$, then

$$(5) \quad e^{-\pi\xi^2} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x \xi} dx.$$

This gives a new proof of the fact that $e^{-\pi x^2}$ is its own Fourier transform, a fact we proved in Theorem 1.4 of Chapter 5 in Book I.

If $\xi = 0$, the formula is precisely the known integral²

$$1 = \int_{-\infty}^{\infty} e^{-\pi x^2} dx.$$

Now suppose that $\xi > 0$, and consider the function $f(z) = e^{-\pi z^2}$, which is entire, and in particular holomorphic in the interior of the toy contour γ_R depicted in Figure 8.

²An alternate derivation follows from the fact that $\Gamma(1/2) = \sqrt{\pi}$, where Γ is the gamma function in Chapter 6.

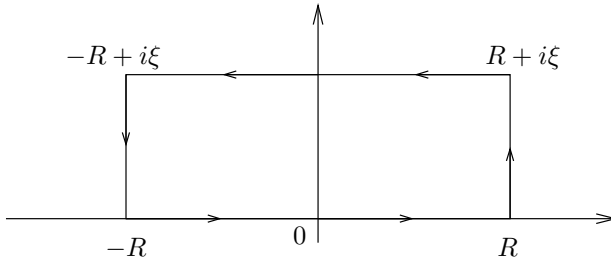


Figure 8. The contour γ_R in Example 1

The contour γ_R consists of a rectangle with vertices $R, R + i\xi, -R + i\xi, -R$ and the positive counterclockwise orientation. By Cauchy's theorem,

$$(6) \quad \int_{\gamma_R} f(z) dz = 0.$$

The integral over the real segment is simply

$$\int_{-R}^R e^{-\pi x^2} dx,$$

which converges to 1 as $R \rightarrow \infty$. The integral on the vertical side on the right is

$$I(R) = \int_0^\xi f(R + iy) i dy = \int_0^\xi e^{-\pi(R^2 + 2iRy - y^2)} i dy.$$

This integral goes to 0 as $R \rightarrow \infty$ since ξ is fixed and we may estimate it by

$$|I(R)| \leq C e^{-\pi R^2}.$$

Similarly, the integral over the vertical segment on the left also goes to 0 as $R \rightarrow \infty$ for the same reasons. Finally, the integral over the horizontal segment on top is

$$\int_R^{-R} e^{-\pi(x+i\xi)^2} dx = -e^{\pi\xi^2} \int_{-R}^R e^{-\pi x^2} e^{-2\pi i x \xi} dx.$$

Therefore, we find in the limit as $R \rightarrow \infty$ that (6) gives

$$0 = 1 - e^{\pi\xi^2} \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x \xi} dx,$$

and our desired formula is established. In the case $\xi < 0$, we then consider the symmetric rectangle, in the lower half-plane.

The technique of shifting the contour of integration, which was used in the previous example, has many other applications. Note that the original integral (5) is taken over the real line, which by an application of Cauchy's theorem is then shifted upwards or downwards (depending on the sign of ξ) in the complex plane.

EXAMPLE 2. Another classical example is

$$\int_0^\infty \frac{1 - \cos x}{x^2} dx = \frac{\pi}{2}.$$

Here we consider the function $f(z) = (1 - e^{iz})/z^2$, and we integrate over the indented semicircle in the upper half-plane positioned on the x -axis, as shown in Figure 9.

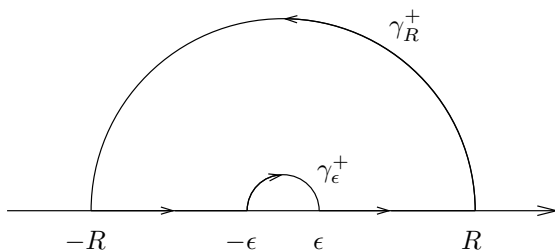


Figure 9. The indented semicircle of Example 2

If we denote by γ_ϵ^+ and γ_R^+ the semicircles of radii ϵ and R with negative and positive orientations respectively, Cauchy's theorem gives

$$\int_{-R}^{-\epsilon} \frac{1 - e^{ix}}{x^2} dx + \int_{\gamma_\epsilon^+} \frac{1 - e^{iz}}{z^2} dz + \int_\epsilon^R \frac{1 - e^{ix}}{x^2} dx + \int_{\gamma_R^+} \frac{1 - e^{iz}}{z^2} dz = 0.$$

First we let $R \rightarrow \infty$ and observe that

$$\left| \frac{1 - e^{iz}}{z^2} \right| \leq \frac{2}{|z|^2},$$

so the integral over γ_R^+ goes to zero. Therefore

$$\int_{|x| \geq \epsilon} \frac{1 - e^{ix}}{x^2} dx = - \int_{\gamma_\epsilon^+} \frac{1 - e^{iz}}{z^2} dz.$$

Next, note that

$$f(z) = \frac{-iz}{z^2} + E(z)$$

where $E(z)$ is bounded as $z \rightarrow 0$, while on γ_ϵ^+ we have $z = \epsilon e^{i\theta}$ and $dz = i\epsilon e^{i\theta} d\theta$. Thus

$$\int_{\gamma_\epsilon^+} \frac{1 - e^{iz}}{z^2} dz \rightarrow \int_\pi^0 (-ii) d\theta = -\pi \quad \text{as } \epsilon \rightarrow 0.$$

Taking real parts then yields

$$\int_{-\infty}^{\infty} \frac{1 - \cos x}{x^2} dx = \pi.$$

Since the integrand is even, the desired formula is proved.

4 Cauchy's integral formulas

Representation formulas, and in particular integral representation formulas, play an important role in mathematics, since they allow us to recover a function on a large set from its behavior on a smaller set. For example, we saw in Book I that a solution of the steady-state heat equation in the disc was completely determined by its boundary values on the circle via a convolution with the Poisson kernel

$$(7) \quad u(r, \theta) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) u(1, \varphi) d\varphi.$$

In the case of holomorphic functions, the situation is analogous, which is not surprising since the real and imaginary parts of a holomorphic function are harmonic.³ Here, we will prove an integral representation formula in a manner that is independent of the theory of harmonic functions. In fact, it is also possible to recover the Poisson integral formula (7) as a consequence of the next theorem (see Exercises 11 and 12).

Theorem 4.1 *Suppose f is holomorphic in an open set that contains the closure of a disc D . If C denotes the boundary circle of this disc with the positive orientation, then*

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta \quad \text{for any point } z \in D.$$

³This fact is an immediate consequence of the Cauchy-Riemann equations. We refer the reader to Exercise 11 in Chapter 1.